GB Auto Fuse & Cable Protection Tool

Engineering Brief for Electrical and Software Teams

Document GBA-0002 · Version 2

GB Engineering

June 2026

Abstract

This document briefs electrical, software, and multidisciplinary engineers on the GB Auto Fuse and Cable Protection web application. It presents the electrical theory and equations implemented in the calculation engine, the software architecture and code logic, traceability to specification GBA-0002, known limitations, and guidance for client-facing communication. The tool replaces a legacy Excel workbook and incomplete MATLAB prototype with a tested TypeScript engine, normalized JSON data libraries, and a mobile-friendly Next.js user interface.

Contents

Chapter 1

Introduction and scope

1.1 Purpose of the tool

Heavy machinery on mine sites and civil fleets typically operates on **24 V** starting circuits. During engine cranking, the starter motor draws hundreds to thousands of amperes for several seconds. The battery, starter cable, and fuse must be coordinated so that:

- the cable carries **continuous** alternator current after start;
- the cable survives **short-time cranking** current without thermal damage;
- voltage drop at the starter remains acceptable;
- the fuse is sized above normal load but withstands cranking without nuisance blowing;
- the fuse **protects** the cable from sustained overcurrent.

The web application automates these checks using fleet data and manufacturer libraries (cable capacity, Littelfuse MEGA 32 V fuses, copper adiabatic constants). Outputs are **decision-support recommendations**—not certified design sign-off.

1.2 Document audience

Table 1.1: Intended readers and focus areas

Role	Use this document to...
Electrical engineer	Verify equations, assumptions, gaps vs. standards
Software engineer	Understand module boundaries, APIs, tests
Project / client lead	Prepare accurate client responses (Chapter 7)
Other engineers	Gain circuit context without reading source code

1.3 Version history

- **Phase 1:** Library mode, core cable/fuse checks, Excel data migration, 12 unit tests.
- **Version 2 (current):** Completeness gate, manual entry, GBA-0002 output panel, fuse-protects-cable check, 26 unit tests.
- **Version 3 (planned):** Parallel fuses, admin UI, PDF export, installation derating.

1.4 Repository layout

```
fuse-tool/
```

```
apps/web/           Next.js 15 + Tailwind UI (client)
packages/engine/    @fuse-tool/engine pure TypeScript (calculations)
data/               JSON libraries exported from Fuse_GUI_APP.xlsx
scripts/            import_from_xlsx.py
docs/               Specifications, user guide, this brief
```

Chapter 2

Electrical theory and circuit context

2.1 Single-line model

The analysed circuit is simplified to:

Battery → Fuse → Cable (out & return) → Starter motor

During cranking, current is dominated by starter torque demand. After start, current falls to alternator charging and vehicle loads. The fuse must protect the cable under fault and overload while not opening during legitimate cranking.

2.2 Typical 24 V boundary conditions

Values align with project documentation and `data/constants.json`:

Table 2.1: Electrical boundary conditions (typical)

Parameter	**Typical value**	**Source**
System voltage V_{sys}	24 V	Machine record or manual entry
Minimum battery voltage V_{min}	16.48 V	Constants (16 V + 3%)
Peak cranking limit I_{limit}	500–1000 A	Per machine `peakCurrentCutoffA`
Cranking current I_{crank}	200–1000 A	Measured / specified
Cranking time t	≥ 5 s floor	$\mathtt{max}(t_{\mathrm{meas}}, 5\,\mathrm{s})$
Alternator continuous I_{alt}	80–150 A	Machine / manual entry
Voltage drop limit	3%	Configurable constant
Fuse safety factor	25%	Configurable (default)

2.3 Applicable standards (reference)

The tool references (but does not automatically enforce full compliance with):

- AS/NZS 5000.1 — electric cables, extruded insulation;
- AS/NZS 1125 — conductors in electric cables;
- AS/NZS 1995:2003 — welding cable duty cycles (reference for cranking duty).

IEC and SAE references appear in the broader project plan for future library expansion. **Engineers must confirm** installation method, grouping, ambient temperature, and OEM requirements beyond what Version 2 calculates.

Chapter 3

Equations and calculation rules

All equations below are implemented in `@fuse-tool/engine`. Each produces a `CheckResult` with status `pass`, `fail`, `warning`, `unavailable`, or `invalid`.

3.1 Battery checks

3.1.1 Cranking voltage

Compare measured voltage during cranking to minimum allowable battery voltage:

$$\text{Pass if } V_{\text{crank}} \geq V_{\text{min}} \tag{3.1}$$

Implemented only when both values are present (library data or manual optional fields). A fail flags battery investigation or replacement.

3.1.2 Cranking time

Compare measured cranking duration to allowed maximum:

$$\text{Pass if } t_{\text{meas}} \leq t_{\text{allowed}} \tag{3.2}$$

3.2 Starter motor cranking current

The engine does **not** calculate cranking current from mechanical power in Version 2. It uses stored or manually entered I_{crank} and compares to the machine peak limit:

$$\text{Pass if } I_{\text{crank}} \leq I_{\text{limit}} \tag{3.3}$$

where I_{limit} is `peakCurrentCutoffA` per vehicle, or `defaultPeakCrankingLimitA` from constants when absent.

GBA-0002 note: The specification describes deriving cranking current from starter motor electrical power:

$$I_{\text{crank}} = \frac{P_{\text{elec}}}{V_{\text{starter}}} = \frac{P_{\text{mech}}/\eta}{V_{\text{starter}}} \tag{3.4}$$

Starter motor library data is imported into machine records but lookup-based calculation is deferred to a future release.

3.3 Cable continuous-current check

The alternator continuous output must not exceed the cable's rated continuous current:

$$\text{Pass if } I_{\text{alt}} \leq I_{\text{cable,cont}} \tag{3.5}$$

Version 2 limitation: No installation derating (ambient, grouping, enclosure) is applied. Ratings are taken as entered or from the fleet record.

3.4 Cable inrush / adiabatic peak capability

For short-duration cranking, copper conductors may carry current above continuous rating using the **adiabatic equation**:

$$I_{\text{allow}} = \frac{K \cdot S}{\sqrt{t}} \tag{3.6}$$

where:

- K = copper constant ($\text{A}\cdot\sqrt{\text{s}}/\text{mm}^2$) from `copper-k-factors.json` by cable type;
- S = conductor cross-section (mm^2);
- t = cranking time used for the check (seconds).

Pass rule:

$$I_{\text{crank}} \leq I_{\text{allow}} \tag{3.7}$$

Required cranking time:

$$t_{\text{req}} = \max(t_{\text{meas}}, t_{\text{min}}), \quad t_{\text{min}} = 5\,\text{s} \tag{3.8}$$

Validity window: If $t >$ `maxAdiabaticCrankingTimeS` (5 s), the engine returns `warning`—the adiabatic model is not intended for extended cranking; investigate mechanical or battery faults.

Legacy correction: The Excel workbook erroneously used $t = 15/1000\,\text{s}$ in $\sqrt{t}$ (cell G13). The engine uses actual cranking time.

Inverse (maximum time for given current):

$$t_{\text{max}} = \left(\frac{K \cdot S}{I_{\text{crank}}}\right)^2 \tag{3.9}$$

3.4.1 Typical K values (copper)

Insulation / type	K (approx.)
Thermoplastic 70 °C PVC	115
Thermoplastic 90 °C PVC	100
Thermosetting 90 °C XLPE	143
Thermosetting 60 °C rubber	141

3.5 Voltage drop

Round-trip resistance (out and return conductors):

$$R_{\text{total}} = 2 \cdot L \cdot \frac{R_{\Omega/\text{km}}}{1000} \tag{3.10}$$

Voltage drop and percentage:

$$\Delta V = I_{\text{crank}} \cdot R_{\text{total}}, \qquad \Delta V\% = \frac{\Delta V}{V_{\text{base}}} \times 100 \tag{3.11}$$

$R_{\Omega/\text{km}}$ is looked up from `cable-capacity.json` by cable size. V_{base} is system voltage or measured cranking voltage when available.

Pass/warning: $\Delta V\% \leq$ voltage drop limit (default 3%). Exceeding the limit yields `warning` in Version 2.

3.6 I²t thermal energy

Thermal energy associated with a current pulse:

$$\mathrm{I}^2 t = I^2 \cdot t \quad [\mathrm{A}^2\mathrm{s}] \tag{3.12}$$

Example: 1000 A for 5 s $\Rightarrow 5 \times 10^6\,\mathrm{A}^2\mathrm{s}$.

Fuse withstand time from I^2t rating (supplementary):

$$t_{\text{fuse}} = \frac{(\mathrm{I}^2 t)_{\text{fuse}}}{I^2} \tag{3.13}$$

The engine uses I^2t as a **supplementary audit check**. Primary fuse cranking validation uses the MEGA 32 V time-current graph.

3.7 Fuse continuous rating (safety factor)

When the cable continuous check passes:

$$I_{\text{fuse,target}} = \left(1 + \frac{s}{100}\right) \cdot I_{\text{alt}} \tag{3.14}$$

Default safety factor $s = 25\% \Rightarrow$ multiply by 1.25.

Example: $I_{\text{alt}} = 389\,\mathrm{A}$, $s = 25\% \Rightarrow I_{\text{fuse,target}} = 486.25\,\mathrm{A}$.

3.8 Fuse library selection

The closest standard rating from `fuse-library.json` minimizes $|I_{\text{rating}} - I_{\text{fuse,target}}|$.

3.9 Fuse time-current withstand (MEGA 32 V)

For breaking current I_{break} (typically peak cranking limit) and selected fuse rating, the engine looks up withstand time $t_{\text{withstand}}$ from `mega32v-curve.json` or fuse record `timeFromGraphS`.

Pass rule:

$$t_{\text{withstand}} \geq t_{\text{req}} \tag{3.15}$$

If insufficient, the engine **escalates** to the next standard fuse rating (up to 8 steps) and re-evaluates.

3.10 Fuse protects cable

The selected fuse must not exceed cable continuous rating (derating not applied in V2):

$$\text{Pass if } I_{\text{fuse,selected}} \le I_{\text{cable,cont}} \tag{3.16}$$

3.11 Parallel fuses (not implemented)

GBA-0002 defines:

$$I_{\text{per fuse}} = \frac{I_{\text{total}}}{n} \tag{3.17}$$

$$I_{\text{derated}} = I_{\text{fuse}} \cdot n \cdot f_{\text{parallel}} \tag{3.18}$$

Parallel fuse logic (Excel G55–G57) is planned for Version 3.

3.12 Overall result aggregation

Each check returns a status. The summary uses the **worst** status in priority order:

`fail` > `invalid` > `warning` > `unavailable` > `pass`

Cable and fuse **suitability badges** on the GBA-0002 output panel aggregate subsets of checks (see `outputs.ts`).

Chapter 4

Data libraries and traceability

4.1 JSON data sources

File	Excel sheet	Content
`machines.json`	MachinesOnSite	Fleet vehicles, cranking data, cable fields
`cable-capacity.json`	Cable_Capacity	Size, continuous I, $R_{\Omega/\mathrm{km}}$
`fuse-library.json`	Fuse_Library	Ratings, $\mathrm{I}^2\mathrm{t}$, part numbers
`mega32v-curve.json`	MEGA32V	Time-current withstand matrix
`copper-k-factors.json`	Copper_K_Factor	Adiabatic K by insulation
`constants.json`	User Input / MD6250	Defaults, standards list

Regenerate from Excel: `npm run import:data` (`scripts/import_from_xlsx.py`).

4.2 Vehicle completeness (Version 2)

Full recommendations require all fields in `REQUIRED_FIELDS` (`completeness.ts`):

- `peakCrankingCurrentA`, `alternatorContinuousA`
- `cableType`, `cableSizeMm2`, `cableContinuousA`, `cableLengthM`, `operatingTempC`
- `peakCurrentCutoffA`, `electricalSystemV`
- Cranking time (measured or required)

As of Version 2, **9 of 37** fleet records are complete. Incomplete records remain visible but `blocked: true` prevents partial unsafe outputs.

Chapter 5

Software architecture

5.1 Design principles

1. **Separation of concerns:** All calculations live in `@fuse-tool/engine`; the web UI only renders results.
2. **Pure functions:** Cable and fuse modules are deterministic given inputs and database snapshots.
3. **Auditable outputs:** Every check includes `specification`, `message`, and optional `legacyReference`.
4. **Fail-safe data handling:** Incomplete library vehicles and invalid manual input are blocked before calculation.
5. **Configuration over hard-coding:** Defaults load from `constants.json`; assumptions listed in `derived.assumptionsUsed`.
6. **Test-driven regression:** Vitest golden vectors (e.g. B45E) and Version 2 acceptance tests.

5.2 Technology stack

Layer	Technology
UI	Next.js 15, React, Tailwind CSS
Engine	TypeScript (ES modules), compiled with `tsc`
Tests	Vitest
Data	JSON (imported from Excel)
Deployment	Vercel / Netlify (static + SSR)

5.3 Calculation pipeline

5.4 Module reference

Module	**Responsibility**
`calculate.ts`	Orchestrator: gates, `runFullCalculation`, `recommend()` wrapper
`cableChecks.ts`	Continuous and adiabatic peak; voltage drop formula
`fuseSelection.ts`	Target rating, closest match, MEGA32V escalation, I^2t, protects-cable

Module	Responsibility
lookups.ts	Machine find, K-factor, resistance, fuse graph lookup
completeness.ts	Required field assessment for library vehicles
validation.ts	Manual entry field validation
outputs.ts	GBA-0002 cable/fuse output block and suitability aggregation
parseValue.ts	Legacy Excel cell parsing (#VALUE!, etc.)
types.ts	Shared interfaces: CheckResult, RecommendationResult
catalog.ts	loadDatabase, listModelIds
recommendLegacyNotes.ts	Documented fixes vs Excel/MATLAB

5.5 Public API

Entry point `calculate(db, request)`:

```
// Library mode
calculate(db, { mode: "library", modelId: "B45E",
  safetyFactorPercent?: number, voltageDropLimitPercent?: number });

// Manual mode
calculate(db, { mode: "manual", inputs: ManualEntryInput });
```

Legacy wrapper: `recommend(db, { modelId })` → library mode only.
Key result fields:

- `blocked` — true if incomplete or invalid input
- `checks[]` — auditable check cards
- `fuse` — target/selected rating, part record, I^2t
- `outputs` — GBA-0002 cable/fuse panel (null when blocked)
- `derived.assumptionsUsed[]` — transparency for engineers
- `implementationNotes[]` — legacy bug fixes applied

5.6 Web application structure

- `Calculator.tsx` — mode tabs, library picker, triggers `calculate()`
- `ManualEntryForm.tsx` — validated manual inputs
- `PdfResultsPanel.tsx` — GBA-0002 required outputs
- `IncompleteVehiclePanel.tsx` — blocked vehicle messaging
- `CheckCard.tsx` — expandable check details
- `lib/db.ts` — loads `bundle.json` into engine

The UI never implements formulas; it only displays engine output. This allows CLI testing, future mobile apps, and API deployment without duplicating logic.

5.7 Testing strategy

Tests load real JSON from `data/` — no mocked cable/fuse tables — ensuring import integrity.

5.8 Legacy corrections applied

Full register: `docs/LEGACY_BUGS_FIXED.md`.

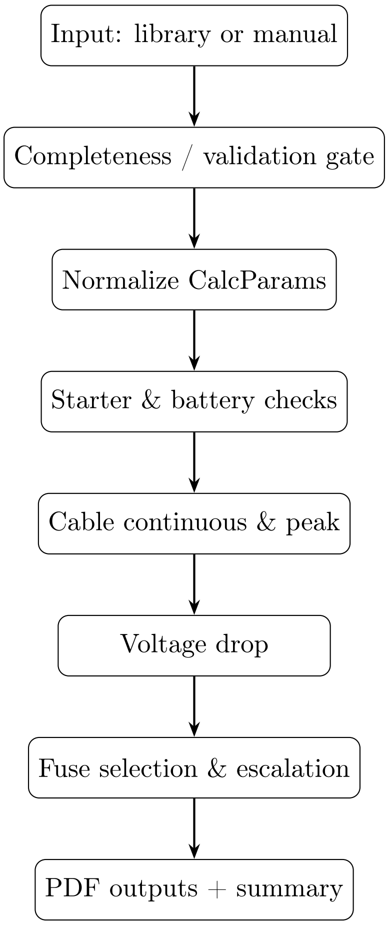

Figure 5.1: High-level calculation flow (`calculate.ts`)

Table 5.2: Test coverage (`npm test` — 26 tests)

Suite	Coverage
`recommend.test.ts`	Golden vectors B45E, D10T, high cranking, unknown model
`v2.test.ts`	Completeness, manual validation, blocked vehicles, fuse-protects-cable

ID	Issue	Engine fix
EX-01	G13 used 0.015 s in $\sqrt{t}$	Use cranking time
EX-02	G31 compared cutoff to capability	Compare I_{crank} to I_{allow}
EX-03	G19 compared strings to voltage	Compare V_{crank} to $V_{\min}$
ML-01	MATLAB truncated columns A:X	Full machine JSON
ML-03	Hard-coded $K = 143$	Lookup table + override

Chapter 6

GBA-0002 compliance matrix

Requirement	**V2**	**V3+**	**Notes**
Machine library selection	✓		37 vehicles
Manual entry mode	✓		Validated inputs
Battery voltage / time checks	✓		When data present
Starter cranking limit	✓		
Cable continuous & inrush	✓		No derating
Voltage drop	✓		
I^2t	✓		Supplementary
Fuse sizing + library match	✓		
MEGA32V withstand + escalation	✓		
Fuse protects cable	✓		Raw ratings only
GBA-0002 output fields (8 items)	✓		
Incomplete vehicle blocking	✓		
Starter motor power → I_{crank}		◦	Uses measured I
Parallel fuse logic		◦	
Cable/fuse temperature derating		◦	
Fault interrupting rating check		◦	
PDF export		◦	On-screen only
Admin fleet editor		◦	JSON/Excel today

Legend: ✓ implemented; ◦ planned.

Chapter 7

Client communication guide

This chapter helps engineers respond accurately to the client (GB Auto / fleet operator) regarding scope, capability, limitations, and deliverables. Use factual language; avoid over-promising certification or standards compliance the tool does not perform automatically.

7.1 Executive summary (client-ready)

We have delivered **Version 2** of the Fuse and Cable Protection web application per specification GBA-0002. The tool analyses 24 V machine starting circuits—battery, cable, starter, and fuse—using your fleet database and manufacturer fuse/cable libraries. It provides pass/fail engineering checks, suggested fuse ratings with part numbers, and a standardized cable/fuse output summary. The application runs in a standard web browser, is mobile-friendly, and separates calculation logic from the user interface for maintainability and testing. **26 automated tests** verify core calculations. The tool is a **design aid**; final selections require qualified engineering review before installation.

7.2 What we delivered

1. **Web application** — vehicle library and manual entry modes.
2. **Calculation engine** — documented equations, auditable check cards, legacy Excel/MATLAB corrections.
3. **Data migration** — Excel workbook exported to version-controlled JSON.
4. **Documentation** — user guide, calculation spec, this engineering brief.
5. **Automated tests** — regression protection for fleet golden cases.

7.3 What to tell the client about data quality

The fleet library contains 37 machines. **Nine** currently have complete engineering data for automated full recommendations. The remainder are visible in the list but clearly marked as incomplete; the application lists missing fields and does not produce partial fuse/cable recommendations that could be misinterpreted. Incomplete cases can be analysed via **Manual entry** once site measurements are available, or fleet records can be updated in the master spreadsheet and re-imported.

7.4 Frequently asked client questions

7.4.1 "Does this replace our Excel tool?"

Response: For the core starting-circuit checks (cable continuous/peak, voltage drop, fuse sizing, MEGA32V withstand), yes—with corrected logic where the legacy workbook had known errors. Features not yet migrated include parallel fuse sizing and starter-motor-derived cranking current. The Excel file remains the master data source until an admin UI is built.

7.4.2 "Is the result compliant with AS/NZS?"

Response: The tool applies standard sizing *logic* referenced in your project documentation but does **not** perform automatic standards certification. Installation derating, grouping, ambient temperature, fault-current interrupting rating, and site-specific limits must be confirmed by a competent person against applicable AS/NZS, IEC, and manufacturer datasheets.

7.4.3 "Why is my machine blocked?"

Response: Version 2 requires complete cable, cranking, alternator, and system voltage data before issuing a full recommendation. This prevents unsafe partial outputs. Use Manual entry with site measurements, or supply missing fields for database update.

7.4.4 "Can we trust the fuse recommendation?"

Response: The suggested fuse is the closest standard rating from the Littelfuse MEGA 32 V library, escalated if the time-current curve requires a higher rating to survive cranking. It includes I^2t cross-check and cable protection verification. Engineering approval is still required—especially where cable continuous margin is tight or fleet data is aged.

7.4.5 "What is still on the roadmap?"

Response: Version 3 priorities include parallel fuse logic, PDF export, browser-based fleet data editing, enhanced cable derating, and optional starter motor library integration for calculated cranking current.

7.5 Suggested disclaimer (include in proposals and UI)

> This calculator follows standard electrical sizing logic for fleet starting circuits but is a design aid only. Final cable and fuse selection must be confirmed against applicable AS/NZS standards, manufacturer datasheets, site requirements, installation conditions, and review by a qualified electrical engineer before implementation.

(This text appears in the web application footer and `docs/USER_GUIDE.md`.)

7.6 Acceptance criteria mapping (GBA-0002 Version 2)

7.7 Escalation paths for engineering review

When presenting results to the client, flag these conditions for human review regardless of pass status:

- Any `fail` on cable continuous, cable peak, or fuse protects cable.
- `warning` on voltage drop or I^2t marginal.

Criterion	Met
Complete vehicles produce cable and fuse recommendations	Yes
Incomplete vehicles visible but safely blocked	Yes
Manual entry validates inputs	Yes
PDF-required outputs displayed	Yes
Logic separated from UI and tested	Yes
README / documentation updated	Yes

- Cranking time $> 5\,\mathrm{s}$ (adiabatic validity).
- Fleet data older than last site measurement.
- Fuse escalation > 0 steps (indicates tight margin on time-current curve).
- Manual entry used because library record incomplete.

Chapter 8

Appendices

8.1 Check ID reference

`id`	Description
`data-completeness`	Library vehicle field gate
`cranking-limit`	$I_{\text{crank}} \leq I_{\text{limit}}$
`battery-voltage`	$V_{\text{crank}} \geq V_{\text{min}}$
`battery-cranking-time`	$t_{\text{meas}} \leq t_{\text{allowed}}$
`cable-continuous`	$I_{\text{alt}} \leq I_{\text{cable}}$
`cable-peak`	Adiabatic peak (Eq. 3.6)
`voltage-drop`	$\Delta V\%$ vs limit
`fuse-rating`	Closest standard fuse
`fuse-withstand`	MEGA32V graph
`fuse-i2t`	Thermal supplementary
`fuse-protects-cable`	Fuse $\leq$ cable rating
`fuse-gb-part`	Part number lookup

8.2 Default constants (`constants.json`)

Constant	Value
`defaultSafetyFactorPercent`	25
`defaultPeakCrankingLimitA`	1000
`minCrankingTimeRequiredS`	5
`maxAdiabaticCrankingTimeS`	5
`defaultKFactorCopper`	143
`defaultElectricalSystemV`	24
`minBatteryVoltage24V`	16.48
`voltageDropPercentLimit`	3

8.3 Related documents

- `docs/USER_GUIDE.md` — field user walkthrough
- `docs/CALCULATION_SPEC.md` — check IDs and modes
- `docs/STANDARDS_AND_CALCULATIONS.md` — standards reference
- `docs/LEGACY_BUGS_FIXED.md` — Excel/MATLAB register

- `README.md` — quick start and API examples

8.4 Building this PDF

From `fuse-tool/docs/engineering/`:

```
pdflatex GBA_FUSE_TOOL_ENGINEERING_BRIEF.tex
pdflatex GBA_FUSE_TOOL_ENGINEERING_BRIEF.tex
```

Run twice for table of contents. Requires a LaTeX distribution (TeX Live, MiKTeX).